

# How the $N = 2, 3, 4$ numerics were actually performed: a full and honest methodology note

Numerical companion to the  $\Sigma$  analytic notes

June 3, 2026

## Abstract

This note documents, in deliberately unvarnished detail, exactly how the numerical study of the RG-invariant entropy-maximal variety  $\Sigma$  was carried out for  $N = 2, 3, 4$  flavours: what equations were solved, how the residual and its Jacobian were built, which solver and acceptance criteria were used, how the solution sets were post-processed into the dimensions and order parameters quoted in `numerics_summary.pdf`, and which independent checks validate each step. It then reflects critically on *why* the numerics cleanly reproduces the analytic loci at  $N = 2$  and  $N = 3$  but reveals a genuinely richer structure at  $N = 4$  — separating the conclusions that are robust from those that are soft, and stating plainly what sampling can and cannot establish. All numbers cited are the ones in `metrics.json`, regenerated by `make_summary.py` from the saved solution sets `n2_clean.npz`, `n3_clean.npz`, `n4_clean.npz` ( $k_{\max} = 3$ ; 196/89/94 solutions).

## 1 What is being computed

The degrees of freedom are the components of a fully symmetric quartic coupling  $\lambda_{ijkl} = \lambda_{(ijkl)}$  on  $\mathbb{R}^N$ , i.e. a point of  $\text{Sym}^4(\mathbb{R}^N)$ , with

$$n_{\text{vars}} = \binom{N+3}{4} = 5, 15, 35 \quad \text{for } N = 2, 3, 4.$$

In code the independent components are indexed by sorted 4-tuples (`make_keys`), and the full tensor is reconstructed by symmetrising over the distinct permutations of each key (`inflate/deflate` in `locus_taylor.py`). We work throughout in the one-loop units  $\epsilon = \kappa = 1$ , in which the beta function is

$$\beta_{ijkl} = -\lambda_{ijkl} + \lambda_{ijmn}\lambda_{mnkl} + \lambda_{ikmn}\lambda_{mnjl} + \lambda_{ilmn}\lambda_{mnjk}. \quad (1)$$

The one-leg  $(1|N)$  reduced density matrix and its spin-2 traceless part are

$$(\rho'_{1|N})_{ij} = \lambda_{iklm}\lambda_{jklm}, \quad T_{ij} = (\rho'_{1|N})_{ij} - \frac{1}{N}(\text{tr}\rho')\delta_{ij}. \quad (2)$$

Entropy of the one-leg cut is maximal precisely on  $\mathcal{M}_{1|N} = \{T = 0\}$ . The object of interest is not  $\mathcal{M}_{1|N}$  itself but the largest RG-flow-invariant subset of it, i.e. the points whose entire forward flow line stays entropy-maximal. Writing the flow derivatives of  $T$  as  $T^{(k)} = (\beta \cdot \partial_\lambda)^k T = k! [t^k] T(\lambda(t))$ , this is the *Lie chain*

$$\Sigma = \{\lambda : T^{(0)} = T^{(1)} = T^{(2)} = \dots = 0\}. \quad (3)$$

Imposing only  $T^{(0)} = 0$  (first-order tangency) is known to be too weak: the three-flavour note (Remark 1) exhibits points that satisfy  $T = T^{(1)} = 0$  yet whose flow leaves  $\mathcal{M}_{1|N}$  at cubic order in RG time,  $\|T(t)\| \sim t^3$ . We therefore solve the *truncated* chain

$$T^{(0)} = T^{(1)} = T^{(2)} = T^{(3)} = 0 \quad (k_{\max} = 3), \quad (4)$$

and separately verify (Section 4) that pushing to  $k_{\max} = 4, 5$  does not shrink the solution set further at sampled points. This  $k_{\max} = 3$  is an *empirical* stabilisation order, not a proven point of ideal-stabilisation; that caveat is carried throughout.

**Why a unit-norm pin is legitimate.** On the variety  $T = 0$  the linear part of (1) acts on  $T$  as a pure rescaling:  $\beta \cdot \partial_\lambda T = -2T + (\text{quadratic-in-}\lambda) \cdot \partial T$ , so on  $\{T = 0\}$  every higher condition  $T^{(k \geq 1)}$  reduces to a *homogeneous* polynomial in  $\lambda$ . Hence  $\Sigma$  is a cone (a union of rays through the origin), and slicing it with the unit sphere  $\|\lambda\| = 1$  captures each ray exactly once. Operationally we append  $\|x\|^2 - 1$  as the final residual; it both removes the trivial root  $\lambda = 0$  and renders the system square (favourable for the Levenberg–Marquardt driver). The reported manifold dimensions are *cone* dimensions (the pin row is dropped before the Jacobian rank is taken).

## 2 How the residual and Jacobian are built

Two independent implementations of  $T^{(0..k_{\max})}(\lambda)$  exist; they agree, which is the first line of defence against coding error.

**(i) Symbolic build (locus\_1to3\_Nflav.py).** Here  $\rho$ ,  $T$ ,  $\beta$  and each Lie derivative  $\mathcal{L}_\beta T = \sum_\alpha \beta_\alpha \partial T / \partial \lambda_\alpha$  are formed symbolically in `sympy`; the residual and an *exact* Jacobian are then `lambdify`'d to NumPy. This is transparent but does not scale: the symbolic expansion of  $(\mathcal{L}_\beta)^k T$  blows up combinatorially and is already painful at  $k = 3$ , so it was used mainly for  $N = 2, 3$  cross-checks and for the  $O(N)$  orbit/PCA diagnostics it also houses (`classify_keys`, `flattening_matrix`).

**(ii) Taylor-mode build (locus\_taylor.py, production).** Rather than expanding  $(\mathcal{L}_\beta)^k T$  symbolically, we integrate the flow as a truncated power series. Each coupling is carried as a degree- $K$  series  $\lambda_\alpha(t) = \sum_{k=0}^K a_k t^k$  (the `Taylor` class: addition, truncated convolution for products, and an `einsum` that contracts the tensor axes coefficient-by-coefficient along the  $t$ -axis). Matching  $t$ -powers in  $\dot{\lambda} = \beta(\lambda)$  gives the recursion

$$(k+1) a_{k+1} = [t^k] \beta(\lambda(t)), \quad k = 0, \dots, K-1, \quad (5)$$

implemented in `evolve_lambda`. Feeding  $\lambda(t)$  through (2) as a series yields  $T(\lambda(t))$ , and the chain values are read off directly as  $T^{(k)} = k! [t^k] T(\lambda(t))$ . Each evaluation costs  $O(N^8)$  for the contractions in (1)–(2) times  $O(K^2)$  for the truncated products — with *no* symbolic blow-up. The residual vector is the concatenation, over  $k = 0, \dots, k_{\max}$ , of the upper-triangular entries of  $T^{(k)}$  *minus* the redundant  $(N-1, N-1)$  entry (fixed by tracelessness), followed by the norm pin.

**Complex-step Jacobian.** The Jacobian fed to the solver is computed by complex-step differentiation: column  $i$  is  $\text{Im } R(x + ih e_i) / h$  with  $h = 10^{-30}$ . Because the residual is evaluated in a dtype-preserving way (the `Taylor/inflate` machinery propagates `complex128` faithfully), this returns the derivative to machine precision with *no* subtractive cancellation — strictly better than a finite difference, and the reason the LM steps are clean even near the rank-deficient solution manifold. (The docstring's mention of a finite-difference Jacobian is stale; the code in `make_residual` uses the complex-step `cs_jacobian`.)

### 3 The solver and acceptance criteria

Solutions are found by *multistart* Levenberg–Marquardt (`scipy.optimize.least_squares`), in `run_taylor.py`:

- **Starts.** Each start  $x_0$  is drawn from a standard normal in  $\mathbb{R}^{n_{\text{vars}}}$  and projected to the unit sphere. The RNG is seeded (default `seed=0`) for reproducibility.
- **Method.** `'lm'` (MINPACK) is used when the residual count  $\geq n_{\text{vars}}$  (true here once the chain and the pin are stacked); otherwise `'trf'`. LM is the right tool because the joint locus is a *positive-dimensional* variety, so the Jacobian at a solution is rank-deficient; the damping term handles that kernel gracefully.
- **Tolerances.** `xtol=ftol=gtol=1e-15`, `max_nfev = 200` (occasionally 400). The tight inner tolerances let LM ride down to the floor of the residual.
- **Acceptance.** A start is *accepted* if  $\max |\text{polynomial residual}| < \text{tol} = 10^{-6}$  and the recovered norm lies in  $(0.5, 2.0)$ ; the accepted point is then rescaled to  $\|x\| = 1$ . The norm window rejects runaway/degenerate fits; the  $10^{-6}$  floor is the nominal membership threshold for  $\Sigma$ .
- **Checkpointing.** Solutions are written to `.npz` every 25 starts so a kill or timeout never loses progress.

The accepted, de-duplicated points are the clean sets `n2/n3/n4_clean.npz`. Acceptance is on  $\|\cdot\|_\infty < 10^{-6}$ , but the *realised* residuals are far tighter for the bulk (see Section 6); the one exception is a handful of  $N = 3$  outliers discussed below.

### 4 Validation: the residual is independently trusted

Because every downstream number rests on the residual being correct, it is checked against machinery that shares no code path (`_reconcile.py`):

- A1 — flow consistency.** The Taylor reconstruction  $T(t) = \sum_k T^{(k)} t^k / k!$  is compared, at several  $t$ , against an *independent* adaptive Runge–Kutta integration (`solve_ivp`, DOP853, `rtol=10^{-12}`) of  $\dot{\lambda} = \beta(\lambda)$  followed by a direct evaluation of  $T$ . Agreement to  $\lesssim 10^{-7}$ . This is the key check: it validates the entire Taylor evaluator against a black-box ODE solve.
- A2/A3 —  $O(N)$  equivariance.** The symmetric ansatz point ( $\lambda_{iiii} = u$ ,  $\lambda_{iiij} = w$ , else 0) kills the chain to  $\sim 10^{-9}$ , and so does a random  $O(3)$  rotation of it — confirming that  $\beta$  and  $T$  are correctly  $O(N)$ -equivariant (residual magnitude invariant under rotation).
- B — dimension reconciliation ( $N = 3$ ).** At a generic locus point the Jacobian nullity is 5; the three  $\mathfrak{so}(3)$  rotation tangents lie in that kernel (relative  $\|Jv\| \sim 10^{-6}$ ) and are rank 3; hence transverse moduli =  $5 - 3 = 2$ , exactly the note’s two parameters  $(u, w)$ . Imposing  $k_{\text{max}} = 4, 5$  leaves the nullity at 5 — the empirical evidence that the chain has stabilised by  $k_{\text{max}} = 3$ .
- C — the solutions are the note’s locus.** On genuine multistart solutions the note’s algebra holds: the spin-2 trace part  $q_{ij} = \lambda_{ijkk} - \frac{1}{N}(\text{tr})\delta \approx 0$ , the harmonic part  $H = \lambda - sD$  is traceless, and its isotropy defect  $C_{ij} = H_{iklm}H_{jklm} - \frac{1}{N}(\text{tr})\delta \approx 0$ .

## 5 Post-processing: how the reported quantities are defined

For each accepted solution (`make_summary.py`):

- **Manifold (cone) dimension** =  $n_{\text{vars}} - \text{rank } J$ , with  $J$  the pin-free Jacobian and rank determined by a *gap* heuristic (`gap_rank`): the largest base-10 gap in the singular-value spectrum below a  $10^{-3}\sigma_1$  floor. This is robust when there is a clean spectral gap and *soft* (can move by one) when there is not — a caveat we flag explicitly.
- **Orbit dimension** = rank of the  $\mathfrak{so}(N)$  rotation tangents at the point (the infinitesimal  $O(N)$  action on  $\lambda$ , deflated to component space). Generic points have full  $\dim O(N) = \binom{N}{2} = 1, 3, 6$ .
- **Moduli** = manifold dim – orbit dim: the genuine physical parameter count after quotienting the flavour rotations.
- **Spin-2 defect**  $\|q\|$  and **isotropy defect** (normalised  $\|C\|$ ): the two order parameters from the analytic proof.
- **$\beta$  norm**: to flag fixed points ( $\|\beta\| < 10^{-6}$ ).
- **$\Sigma$ -orbit spectrum match**: whether the eigenvalue multiplicities of the  $\text{Sym}^2$  flattening matrix  $M_{(ij)(kl)} = \lambda_{ijkl}$  match the symmetric line's  $[1, N-1, \frac{N(N-1)}{2}]$ . Because this spectrum is an  $O(N)$  invariant, it is a *rotation-proof* test of whether a point is in the  $O(N)$  orbit of the symmetric ansatz.

A PCA of the centred solution cloud is also reported, but it measures the *linear span* of the sampled points, not a manifold dimension, and is treated only as a coarse sanity check (it overshoots the local dimension whenever the union of  $O(N)$  orbits sweeps out extra linear directions).

## 6 Results, case by case

| $N$ | $n_{\text{vars}}$ | #sols | manifold dim (count)     | median | $\max  T^{(0-3)} _{\Sigma}$ | $\Sigma$ -orbit |
|-----|-------------------|-------|--------------------------|--------|-----------------------------|-----------------|
| 2   | 5                 | 196   | 3 (196)                  | 3      | 0                           | 196/196         |
| 3   | 15                | 89    | 5 (86), 10 (3)           | 5      | $1.4 \times 10^{-12}$       | 0/89            |
| 4   | 35                | 94    | 9 (27), 11 (18), 14 (49) | 14     | $6.8 \times 10^{-13}$       | 0/94            |

$N = 2$  (5 couplings, 196 solutions). Residuals reach  $1.4 \times 10^{-12}$ . Every solution has manifold dim 3, orbit dim 1 (=  $\dim O(2)$ ), and moduli 2; the spin-2 defect is  $\|q\| \leq 10^{-15}$  and the isotropy defect vanishes for all 196. Decoding the components as  $(a, b, c, d, e)$ , the dedicated check (`report_N2`) finds  $\max |a - e|$  and  $\max |b + d|$  both at the  $10^{-12}$  level: the numerics lands *exactly* on the analytic odd-block locus  $\mathcal{E}_- = \{(a, b, c, d, e) = (a, b, c, -b, a)\}$  of the two-flavour note (its eq. 43). No fixed points are hit ( $\|\beta\| \in [1.23, 12.71]$ ), as expected for random multistart on a positive-dimensional variety.

$N = 3$  (15 couplings, 89 solutions). The dominant stratum (86/89) has manifold dim 5, orbit dim 3, moduli 2, with  $q = 0$  and isotropic ( $C = 0$ ) on all 86 — precisely the analytic cubic/tetrahedral structure

$$\underbrace{5}_{\text{embedded cone}} = \underbrace{2}_{\text{moduli } (u,w)} + \underbrace{3}_{\dim O(3)},$$

reproducing the three-flavour theorem that  $\Sigma$  is the  $O(3)$  orbit of the two-parameter plane  $\lambda_{iiii} = u$ ,  $\lambda_{ijjj} = w$ , with the spin-2 part *forced* to vanish (Step 1) and a unique isotropic harmonic orbit (Step 2). The  $\Sigma$ -orbit count is 0/89 because the *generic* solution is the broader cubic harmonic, not the special  $SO(3)$  (symmetric-line) point — consistent, not contradictory. The remaining 3/89 are flagged as a “dim 10” stratum with  $q \neq 0$  up to  $6 \times 10^{-3}$  and the loosest residuals in the whole study ( $\leq 3.5 \times 10^{-7}$ ). Given the analytic proof that  $N = 3$  has *exactly one* orbit, the honest reading is that these three are under-converged near-solutions, not a second component: at slightly elevated residual the gap-rank heuristic loses its spectral gap and reports a spuriously low rank (hence a spuriously high dimension). They are a numerical artefact of finite tolerance, fully consistent with  $N = 3$  uniqueness.

**$N = 4$  (35 couplings, 94 solutions).** Residuals are tight ( $\leq 6.1 \times 10^{-12}$ ) and again no fixed points are hit ( $\|\beta\| \in [1.50, 12.32]$ ). But the locus is now *reducible*: the gap-rank manifold dimension takes three values,

| dim | #sols | orbit | moduli | $q=0$ | $\max\ q\ $ | iso $C=0$ |
|-----|-------|-------|--------|-------|-------------|-----------|
| 9   | 27    | 6     | 3      | 5/27  | 0.341       | 5/27      |
| 11  | 18    | 6     | 5      | 0/18  | 0.479       | 0/18      |
| 14  | 49    | 6     | 8      | 49/49 | 0           | 49/49     |

The decisive, threshold-independent finding is the last-but-two column: the spin-2 part  $q$  is *not* forced to zero. 40/94 solutions carry  $q \neq 0$ , up to  $\|q\| = 0.479$  — far above any tolerance, so this is real, not noise. The dim-14 stratum (49 solutions) is the four-flavour analogue of the  $N = 3$  family ( $q = 0$ , isotropic); the dim-9 and dim-11 strata are genuinely different, with large  $q$  and non-isotropic harmonic parts. The symmetric ansatz still solves the chain ( $6.8 \times 10^{-13}$ ), but the  $\Sigma$ -orbit match is 0/94: the locus is strictly larger than the symmetric line’s orbit, and contains inequivalent harmonic-isotropy strata.

## 7 Reflection: clean match at $N = 2, 3$ , genuine break at $N = 4$

**Why  $N = 2, 3$  match.** At  $N = 2$  and  $N = 3$  the analytic loci are not merely sampled but *proven complete*:  $\mathcal{E}_-$  at  $N = 2$ , and at  $N = 3$  the exhaustive solve of the five harmonic-isotropy quadratics  $C_{ij} = 0$  on the nine-dimensional space of harmonic tensors, which yields a single  $O(3)$  orbit (the cubic harmonic). The numerics reproduces both at the level of *independent* corroboration: the order parameters the proof predicts to vanish do vanish on essentially every sampled point ( $q = 0$  and  $C = 0$  on 196/196 at  $N = 2$  and 86/86 on the main  $N = 3$  stratum), the dimension arithmetic  $5 = 2 + 3$  comes out of the Jacobian rank, and the flow-consistency check A1 ties the residual to a black-box ODE integration. The agreement is strong precisely because the analytic side is a theorem and the numerical side is a methodologically independent confirmation of it.

**Why  $N = 4$  is different.** The “discrepancy” at  $N = 4$  is *not* numerics disagreeing with analytics. It is that the clean  $N \leq 3$  paradigm —  $q$  forced to zero, a single isotropic harmonic orbit — *fails*, and both the numerics and the four-flavour analytic note agree that it fails. Analytically, the strictly analogous harmonic computation at  $N = 4$  produces a *second* orbit, the hypertetrahedral  $S_5 \times \mathbb{Z}_2$  fixed point, distinguished from the linear family  $\Sigma_4^{\text{lin}}$  by its  $\text{Sym}^2$ -flattening eigenvalue spectrum (an  $O(4)$  invariant, hence not removable by any rotation):  $\Sigma$  at  $N = 4$  is *strictly larger* than  $\Sigma_4^{\text{lin}}$ . Numerically, the two faces of this are (i)  $q \neq 0$  is now allowed — the  $N = 3$  Step-1 theorem has no

$N = 4$  counterpart — and (ii) the solution set splits into a  $q = 0$  isotropic stratum plus inequivalent  $q \neq 0$  strata. The numerics independently discovers the same qualitative break the analytic note predicts.

**What is robust, and what is soft.** Being honest about the strength of each  $N = 4$  claim:

- **Robust.** The existence of solutions with  $q \neq 0$  ( $\|q\|$  up to 0.479, orders of magnitude above tolerance) is threshold-independent and decisively breaks the  $N = 3$   $q = 0$  theorem at  $N = 4$ . The existence of at least one  $q = 0$ /isotropic stratum (the dim-14 family) alongside genuinely distinct  $q \neq 0$  strata is equally robust. The tight residuals ( $\leq 6 \times 10^{-12}$ ) confirm these are bona fide chain solutions, not near-misses.
- **Soft.** The *fine* stratification — the exact dimensions 9/11/14 and their counts 27/18/49 — rests on the gap-rank heuristic, which can shift a dimension by one on near-degenerate spectra, and on a sample of only 94 accepted points. The word “component” here means “stratum by local Jacobian rank,” not a proven irreducible component.
- **Sampling limits.**  $N = 4$  convergence is hard ( $\sim 30$ – $40\%$  of starts succeed), so rare strata may be under-sampled or missed entirely. Multistart establishes *existence* of points with given properties and *estimates* local dimensions; it cannot prove the stratification is exhaustive, nor prove maximality/completeness of  $\Sigma$ . The analytic note supplies the existence and rotation-inequivalence of the hypertetrahedral component; the numerics cannot, by itself, certify there are no further pieces.

**A note on fixed points.** No solution in any case sits on an RG fixed point ( $\|\beta\| > 1$  throughout). This is expected: the catalogued fixed points (Gaussian, Decoupled, Cubic, Heisenberg at  $N = 3$ ; additionally the hypertetrahedral point at  $N = 4$ ) lie on  $\Sigma$  but form a positive-codimension subset, which random multistart on a continuous variety hits with probability zero. The analytic notes locate them exactly; the numerics samples the generic flowing part of the same variety. There is no tension here.

## 8 Reproducibility and honest limitations

Every number in `numerics_summary.pdf` and in this note is computed by `make_summary.py` from the saved `.npz` sets and dumped to `metrics.json`; nothing is hand-typed. To regenerate: run `run_taylor.py` with the desired  $N$ ,  $k_{\max} = 3$ , and a fixed seed to produce the solution set, then `make_summary.py` to recompute the diagnostics. The principal limitations, restated plainly: (i) the chain is truncated at  $k_{\max} = 3$  and its stabilisation is checked only at sampled points, not proved; (ii) “dimension” is a gap-based local Jacobian rank,  $\pm 1$  on degenerate spectra; (iii) “component/stratum” is not a proven irreducible component; (iv)  $N = 4$  is under-sampled relative to its complexity, so absence of a stratum in the data is not evidence of its absence in  $\Sigma$ ; (v) multistart proves existence and estimates dimension, never completeness. With those boundaries drawn, the numerics does what it is meant to do: it independently confirms the  $N = 2, 3$  theorems and independently exposes the  $N = 4$  break that the four-flavour analytic note then explains.